

SOP-10: 5-Azacytidine (PHS)

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 209
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
5-Azacytidine, 99% (1 g, AC226620000)	320-67-2

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Health hazard (carcinogen, mutagen, reproductive toxicity); Skull and crossbones; Exclamation mark

- PARTICULARLY HAZARDOUS SUBSTANCE. 5-Azacytidine is a cytotoxic antineoplastic agent (a DNA methyltransferase inhibitor and nucleoside analogue) that is mutagenic, clastogenic, embryotoxic and a suspect human carcinogen. MyChem classifies it under 'Other Health Hazards - Including Carcinogens'. It must be handled as a hazardous drug in accordance with NIOSH hazardous drug handling guidance. Sections 9-11 of this SOP are mandatory.
- Stock chemicals: 1 g of 5-azacytidine POWDER. Intermediates and working solutions: concentrated stock solutions in DMSO or in acetic acid/water, and dilute working solutions in culture medium or filtered seawater used to treat cells or larvae in epigenetics experiments. Wastes: unused powder, stock and working solutions, treated culture medium, and contaminated labware and debris.
- The powder is the dominant hazard. Weighing produces a respirable dust of a cytotoxic, mutagenic drug. Because 5-azacytidine is routinely dissolved in DMSO, which drives solutes through intact skin (see SOP-02), a DMSO stock of this compound must be treated as capable of delivering a systemic dose on skin contact.

- Routes of exposure: inhalation of powder, skin contact and absorption (greatly enhanced by DMSO), eye contact, ingestion, and accidental injection.
- How exposure might occur: weighing powder; preparing stock solution; pipetting stock into medium; dosing culture vessels or larval containers; handling treated medium and vessels; disposing of treated waste; and contact with contaminated gloves or surfaces.
- Target organs: bone marrow (myelosuppression), gastrointestinal tract, liver, kidney, and the reproductive system and developing fetus. The compound incorporates into DNA and RNA, so all rapidly dividing tissue is a target.
- Signs and symptoms of exposure: there are typically no acute signs from an incidental occupational exposure; the hazards are genotoxic, reproductive and carcinogenic. Direct contact may cause skin and eye irritation. The absence of symptoms is not evidence that an exposure was harmless.
- GHS classification (from MyChem / SDS): MyChem records 'Fire Class: Other Health Hazards - Including Carcinogens' with GHS categories not populated. Supplier SDSs classify 5-azacytidine variously as Acute toxicity Category 3-4 (oral), Carcinogenicity Category 1B or 2, Germ cell mutagenicity Category 1B or 2, and Reproductive toxicity Category 1B or 2. The most protective classification is used for this SOP.
- No OSHA PEL, WISHA PEL or ACGIH TLV has been established. Control is by containment, designated area and work practice, with the goal of no measurable exposure.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- ALL handling of 5-azacytidine powder and all preparation of stock solutions is performed inside the certified chemical fume hood in FTR 209, on a disposable plastic-backed absorbent liner, with the analytical balance placed inside the hood and the sash at or below the marked working height. Any chemical fume hood used must be tested and passed by EH&S. A Class II biological safety cabinet is NOT an acceptable substitute for powder weighing unless it is a Class II Type B2 vented to the outside; the chemical fume hood is used.
- Dosing of cells or larvae with dilute working solution is performed inside the biological safety cabinet or the fume hood, over an absorbent liner, in closed or covered vessels wherever possible.
- Closed-tube handling of capped stock aliquots (transport, freezing, thawing) may be done at the bench in a closed secondary container.
- Aerosol-generating steps (vortexing, sonicating, pipetting forcefully) are performed inside the hood with the tube capped where possible.
- The designated area is kept clear of unrelated equipment and is covered with a disposable liner that is changed each session.
- The nearest eyewash and safety shower are located in FTR 209 and are inspected and flushed weekly by lab staff. A hazardous drug spill kit (or the general chemical spill kit plus plastic-backed absorbent, a plastic scoop, a puncture-resistant container, and heavy chemotherapy-rated gloves) is maintained in FTR 209.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: a disposable, low-permeability, back-closing gown with knit cuffs (a chemotherapy-rated gown) is worn OVER the standard lab coat for all powder handling and stock preparation, and is discarded as hazardous solid waste after each session. A standard buttoned barrier lab coat plus disposable chemically resistant sleeves is the minimum for dosing with dilute working solutions.
- A chemically resistant apron and face shield are additionally required for spill cleanup.
- Contaminated gowns, sleeves and lab coats are discarded as hazardous waste, not laundered, and never taken home.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: DOUBLE gloves at all times. Use ASTM D6978 chemotherapy-rated nitrile gloves (e.g., Halyard Purple Nitrile-Xtra, Ansell Microflex 93-260) as both layers, with the outer glove over the gown cuff. If the stock solution is prepared in DMSO, the outer glove must be butyl rubber or laminate, because DMSO permeates nitrile rapidly and will carry the drug through it (see SOP-02).
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with the specific chemical being used. Gloves must be inspected prior to use, including a check for pinholes.
- Change the outer glove immediately on any contact and at least every 30 minutes; change both gloves after any spill and at the end of each task.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these chemicals. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 chemical splash goggles for all handling of powder, stock solutions and working solutions. A face shield worn over goggles is required for spill cleanup and for any operation with a splash risk.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP, provided all powder handling is confined to the certified chemical fume hood as written.
- 5-Azacytidine POWDER is NEVER permitted to be handled outside of the chemical fume hood. Dilute working solutions in medium or seawater may be handled in the biological safety cabinet or at the bench in closed vessels.
- If a task cannot be performed in the fume hood, work must stop and the PI must contact EH&S. NIOSH recommends respiratory protection for hazardous drug handling that cannot be contained; that determination is made by EH&S, not in the lab.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit (AL) or Occupational Exposure Limit (OEL) has been exceeded or when there is a possibility that an AL/OEL will be exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program so a respiratory protection analysis can be performed. Program enrollment includes medical evaluation, training and fit testing for an appropriate respirator.

- Where air-purifying respirators are appropriate, use a full-face respirator with a P100 particulate filter (plus organic vapor cartridge if DMSO is involved) as a backup to engineering controls. Use a full-face supplied air respirator if it is the sole means of protection.

Section 4 - Special Handling and Storage Requirements

- Quantity limits: no more than 50 mg of powder is weighed in a single operation. Stock solutions are prepared at the smallest practical volume and aliquoted into single-use tubes so that the powder container and the concentrated stock are opened as few times as possible.
- Substitution and minimization: before each new experiment, consider whether a pre-made solution can be purchased instead of weighing powder, and whether a lower dose or smaller experimental scale would answer the question.
- A dedicated set of consumables is used: disposable weigh boats, disposable spatulas, a dedicated pipette set labelled '5-AZA ONLY', and dedicated tip boxes kept within the designated area.
- All containers - powder bottle, stock tubes, working solution bottles, treated culture vessels and waste containers - must be labeled with the full chemical name, concentration, solvent, the words 'CYTOTOXIC - PARTICULARLY HAZARDOUS SUBSTANCE - HANDLE WITH CHEMOTHERAPY PRECAUTIONS', the date and the preparer's initials.
- Work is performed only during normal business hours when at least one other person is present in or immediately adjacent to FTR 209. Unattended operations with open 5-azacytidine are prohibited; treated cultures incubating in closed, labeled vessels are permitted and the incubator is posted with a cytotoxic hazard label.
- A 'CYTOTOXIC DRUG IN USE - PARTICULARLY HAZARDOUS SUBSTANCE - 5-AZACYTIDINE' sign is posted on the fume hood sash while work is in progress.
- Storage: the powder is stored in its original container inside a labeled, closed, rigid secondary container in the -20 degC freezer in FTR 209, posted with a cytotoxic hazard label. Stock aliquots are stored in a separate labeled closed secondary container in the same freezer. Segregate from food-grade or unrelated reagents; nothing else is stored inside the secondary container.
- Transport: the powder container is not transported outside FTR 209. Aliquots are moved only in closed, leak-proof, labeled secondary containers.
- Clean the chemical fume hood upon completion of tasks: remove and bag the absorbent liner as hazardous waste, then wipe the work surface with detergent solution, followed by clean water, followed by 70% ethanol; wipe dry.
- Clean all contaminated surfaces with detergent solution followed by clean water and 70% ethanol, and dry.
- Place all contaminated disposable items in the dedicated, closed, labeled cytotoxic solid waste container.
- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with 5-azacytidine must be decontaminated at the end of the laboratory work session. Complete this inside the chemical fume hood before removing any of the items. Wherever possible, use disposable labware and discard it rather than reusing it.
- When work is completed, remove the outer glove, then the gown, then the inner glove, and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- POWDER spill of any size: do NOT brush, wipe dry, or use compressed air. If the spill is contained inside the operating fume hood and is less than approximately 50 mg, trained and authorized staff may clean it up: leave the hood running

with the sash low; dampen the powder with water or 70% ethanol using a wash bottle; wipe up with plastic-backed absorbent working from the outside inward; place all material in the cytotoxic solid waste container; then wash the surface three times with detergent solution followed by clean water. A powder spill OUTSIDE the fume hood, or larger than approximately 50 mg: evacuate the room, close the door, post 'DO NOT ENTER - CYTOTOXIC SPILL', prevent entry, and call 9-1-1. Notify EH&S.

- LIQUID spill - less than 25 mL of dilute working solution inside the hood or biological safety cabinet: may be handled by trained and authorized staff wearing goggles, double chemotherapy-rated gloves, gown and sleeves. Absorb with plastic-backed absorbent from the outside inward, place in the cytotoxic solid waste container, and wash the area three times with detergent solution and clean water.
- Any spill of concentrated stock (especially a DMSO stock), any liquid spill greater than 25 mL, any spill outside the designated area, and any spill in which a person may have been exposed: DO NOT clean up. Restrict access, post a sign, and call EH&S at 206-543-0467 during business hours or 9-1-1 after hours. A DMSO stock spill on skin is a systemic exposure, not a surface contamination event.
- PPE required for cleanup: chemical splash goggles AND face shield, double ASTM D6978 chemotherapy-rated gloves (butyl or laminate outer glove if DMSO is involved), disposable chemotherapy gown over the lab coat, disposable sleeves, shoe covers, full-length pants and close-toe shoes.
- Signage and entry restrictions: post 'DO NOT ENTER - CYTOTOXIC SPILL' on the door for any spill outside the containment device; restrict entry until EH&S clears the space.
- Spill area must be cleaned up in the following manner: after absorbent removal, wash the area THREE times with detergent solution followed by clean water, then wipe with 70% ethanol and dry. A single wipe is not adequate for a cytotoxic drug.
- Spill cleanup materials must be disposed of in the following manner: place all absorbent, gloves, gown and disposables in the labeled cytotoxic solid waste container, double bagged, label with a UW Hazardous Waste Label listing '5-azacytidine - cytotoxic - spill debris', and submit a collection request to EH&S promptly.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Skin exposure: remove contaminated gloves, gown and clothing immediately and wash the affected skin with soap and copious running water for at least 15 minutes, using the safety shower for anything more than a small splash. If the material was dissolved in DMSO, treat this as a systemic exposure: flush for 15 minutes and call 9-1-1, stating that a cytotoxic drug in DMSO contacted the skin. Do not scrub.
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open, then go to the Emergency Department.
- Inhalation exposure (powder): move the person out of the contaminated area to fresh air and call 9-1-1.
- Ingestion: rinse mouth with water; do not induce vomiting; call 9-1-1.
- Sharps injury or accidental injection: this is a medical emergency. Scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap and call 9-1-1 immediately.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent (5-azacytidine - state that it is a cytotoxic antineoplastic agent and name the solvent), dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.

- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.
- Every exposure to 5-azacytidine, however small it appears, is reported to the PI and to EH&S so that follow-up can be arranged; there are no acute symptoms to rely on.

Section 6 - Waste Accumulation and Disposal Procedures

- ALL material that has contacted 5-azacytidine is hazardous waste. This includes unused powder, stock and working solutions, treated culture medium and seawater, culture vessels, pipettes and tips, gloves, gowns, absorbent liners and weigh boats.
- Do NOT pour any 5-azacytidine-containing liquid down the drain in any concentration. Do NOT place any contaminated solid in the regular trash. Do NOT autoclave 5-azacytidine waste - autoclaving does not reliably destroy it and can aerosolize it; never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- If treated cultures also contain biological material, the waste is BOTH a chemical and a biological hazard. Chemical hazard takes precedence for disposal routing: do not autoclave. Contact EH&S for guidance on mixed chemical-biological waste before generating it.
- Accumulate liquid waste at the point of generation - inside the fume hood or biological safety cabinet - in a sturdy, chemically compatible HDPE container with a securely closable screw-top lid, kept closed except when actively adding waste, in secondary containment, labeled 'CYTOTOXIC'.
- Solid waste is accumulated in a dedicated, rigid, lined, closed container labeled '5-AZACYTIDINE CYTOTOXIC SOLID WASTE' kept inside the designated area. Sharps contaminated with the drug go into a labeled puncture-resistant container designated for cytotoxic sharps.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage, starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- No neutralization or in-lab treatment of 5-azacytidine waste is authorized.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University, all its campuses, and off-site locations; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Because a 1 g container of a cytotoxic drug is a long-term inventory liability, the PI should review annually whether the remaining material is still needed and surplus it through EH&S if not.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials. Update the MyChem inventory when containers are emptied, received, transferred or surplus.

Section 7 - Protocol

- Procedure A - Weighing 5-azacytidine and preparing stock solution (authorized personnel only). Don a standard lab coat, then a disposable chemotherapy gown, then double chemotherapy-rated gloves. Inside the certified fume hood, with the balance in the hood and a plastic-backed absorbent liner in place: bring the powder container to room temperature inside a closed secondary container before opening (to avoid condensation and caking), open it inside the hood, weigh the required amount into a disposable weigh boat with a disposable spatula, and transfer to a labeled tube. Recap and re-bag the powder container immediately and return it to the freezer secondary container. Add the calculated volume of solvent (DMSO per SOP-02, or acetic acid/water) inside the hood, cap, and vortex to dissolve

with the tube capped. Aliquot into single-use labeled tubes. Discard the weigh boat, spatula, tips and liner into the cytotoxic solid waste container. Wipe the balance and hood three times with detergent solution and clean water, then 70% ethanol.

- Procedure B - Preparing dilute working solution. Inside the fume hood or biological safety cabinet, over an absorbent liner, thaw one single-use aliquot, add it to pre-measured medium or filtered seawater in a capped vessel, and mix by inversion. Label the vessel. Discard the empty aliquot tube and tip into the cytotoxic solid waste container.
- Procedure C - Treating cells or larvae. Inside the biological safety cabinet or fume hood, over an absorbent liner, add working solution to the culture vessel and cap or cover it. Wipe the exterior of the vessel with 70% ethanol before moving it to the incubator. Post the incubator with a cytotoxic hazard label listing the compound, PI and date.
- Procedure D - Harvest and disposal of treated cultures. Inside the fume hood or biological safety cabinet, over an absorbent liner, decant treated medium directly into the labeled cytotoxic liquid waste container. Place vessels, pipettes and tips into the cytotoxic solid waste container. Do not rinse treated vessels into a sink.
- Prohibited: weighing powder outside the fume hood; handling without double gloves and a gown; drain disposal; autoclaving; regular-trash disposal of any contaminated item; use by untrained or unauthorized personnel.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs).
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

Yes, potentially. If 5-azacytidine is administered to live animals - including immersion exposure of bivalve larvae or juveniles, or injection of finfish - the use will be documented and approved by IACUC before the work begins, and the approved protocol number recorded here. Personnel must protect themselves from contaminated animals, tank water and animal waste for at least 48 hours after the last dose: wear double chemotherapy-rated gloves, a disposable gown, safety goggles and a lab coat when handling exposed animals or their containers; keep exposure tanks in a posted, restricted area with secondary containment; suppress aerosols by covering tanks and avoiding vigorous aeration; and collect all tank water, bedding, carcasses and cleaning materials as cytotoxic hazardous waste rather than discharging them to the drain or placing them in regular trash. Tanks and nets used for exposed animals are dedicated to that purpose, labeled, and decontaminated per Section 10 before any other use.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 9 - Approvals Required

- All staff working with 5-azacytidine must be trained on this SOP prior to starting work. They must also review the 5-azacytidine SDS, and it must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- 5-Azacytidine is a particularly hazardous substance and a hazardous drug. SPECIFIC WRITTEN AUTHORIZATION from the Principal Investigator is required before any individual may work with it, and that authorization is recorded on the training roster in Section 12 with the PI's countersignature. Authorization is granted per person, and is required in addition to general lab access.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP and the SDS; (4) documented instruction on hazardous drug handling technique - double gloving, gowning, don/doff sequence,

designated area discipline, and cytotoxic waste segregation; (5) SOP-02 (DMSO) training if a DMSO stock will be prepared.

- A worker must complete at least two supervised runs of Procedures A-D, observed and signed off by the PI, before performing 5-azacytidine work independently for the first time.
- Personnel who are pregnant, planning a pregnancy, breastfeeding, or actively trying to conceive should not handle 5-azacytidine. Any such individual should consult the EH&S Reproductive Health in the Workplace resources and their healthcare provider, and should discuss alternative task assignment with the PI. Such consultation is voluntary and confidential; no disclosure of the reason is required to be reassigned.
- The PI must notify EH&S before beginning any new project involving 5-azacytidine so that the hazardous drug handling controls can be reviewed, and before any animal administration so that IACUC and EH&S requirements are aligned.
- No medical examination is required for the work described because respirators are not used. A medical examination and fit testing would be required before any respirator use.

Section 10 - Decontamination

- Decontamination for a cytotoxic drug requires repeated washing, not a single wipe. The standard cycle is: detergent solution, clean water, detergent solution, clean water, then 70% ethanol, allowing each to dwell before wiping.
- Fume hood and balance: remove and bag the absorbent liner as cytotoxic solid waste, then run the standard cycle above on the work surface, sash interior and balance pan and housing.
- Biological safety cabinet: remove the liner, run the standard cycle on the work surface and interior walls, and record the decontamination in the cabinet log.
- Equipment: the dedicated '5-AZA ONLY' pipettes are wiped through the standard cycle and remain in the designated area. Culture vessels, tubes and tips are disposable and are discarded rather than decontaminated. If reusable glassware must be used, rinse it three times inside the hood into the cytotoxic liquid waste container, then run it through the standard cycle and a full detergent wash.
- Incubators and refrigerators that have held treated cultures are wiped through the standard cycle when the experiment ends, and the cytotoxic hazard label is removed only after that decontamination.
- Animal exposure equipment (tanks, nets, tubing), if used: drain into cytotoxic liquid waste, then run the standard cycle and rinse three times with clean water before the equipment is returned to general use.
- Controlled areas: there are no glove boxes or restricted-access hoods in FTR 209; the designated areas in Section 11 are the controlled areas for this SOP and are decontaminated at the end of each session and monthly regardless of use.
- Cleaning solutions and materials: laboratory detergent (Alconox or Contrad) solution, clean water, 70% ethanol, lint-free wipes, plastic-backed disposable absorbent liners, and a plastic scoop reserved for this purpose.

Section 11 - Designated Area

- DESIGNATED AREA 1 - powder weighing and stock preparation: the certified chemical fume hood in FTR 209, with the analytical balance inside the hood and a plastic-backed absorbent liner in place.
- DESIGNATED AREA 2 - working solution preparation, dosing, harvest and treated-culture handling: the biological safety cabinet and the immediately adjacent bench in FTR 209, over an absorbent liner.
- DESIGNATED AREA 3 - if animal exposures are conducted: the posted exposure tank area, with secondary containment, restricted access, and dedicated tanks, nets and tubing.

- All designated areas are posted with a 'DESIGNATED AREA - PARTICULARLY HAZARDOUS SUBSTANCE - CYTOTOXIC DRUG - 5-AZACYTIDINE - AUTHORIZED PERSONNEL ONLY' sign listing the PI, contact number, required PPE and the authorization requirement.
- Cytotoxic liquid and solid waste containers are kept within the designated areas in secondary containment.
- Storage: labeled, closed, rigid secondary containers in the -20 degC freezer in FTR 209, posted with a cytotoxic hazard label.
- No other bench, room or piece of equipment in the Roberts Lab space is authorized for 5-azacytidine use.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.
- Because this SOP covers a particularly hazardous substance, the Principal Investigator countersigns each entry below to record the written authorization required by Section 9.

I have read and understand the content of this SOP:

[illegible]

Name (print)	Signature	Date